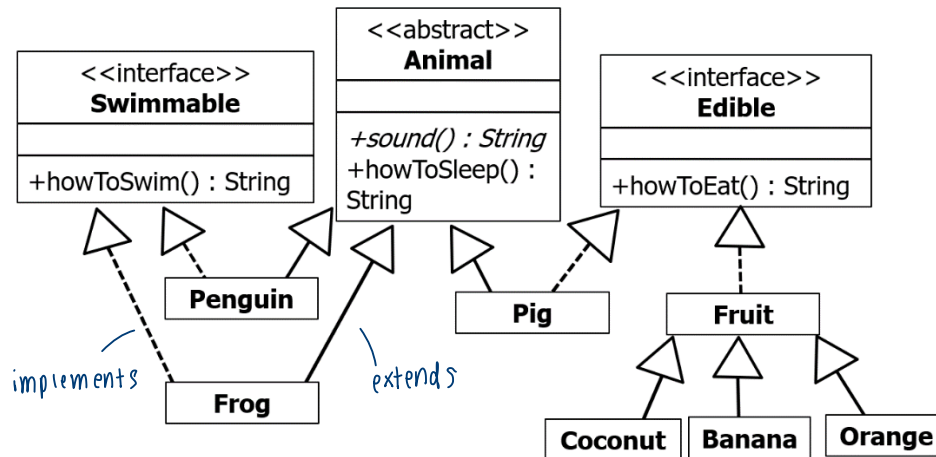




Quiz #08 Interface (27/04/2023)

1. จงเขียนโปรแกรมจาก UML ของคลาสที่ให้มาพร้อมทั้งทำการทดสอบการ **Inheritance** การ **Interface** และการ **Overriding** ของคลาสลูกแต่ละตัวในโปรแกรมทดสอบ เพื่อจะแสดงผลลัพธ์ที่ได้ดังแสดงในตัวอย่าง



โดยโปรแกรมทดสอบจะทำการสร้างอ็อบเจ็กต์และเรียกใช้เมธอดเพื่อแสดงผลลัพธ์ที่ได้บนหน้าจอตามข้อความตัวอย่างนี้

ผลลัพธ์ที่จะแสดงบนหน้าจอ

Pig: Oink-Oink	}	Sound
Penguin: Honk-Honk		
Frog: Ribbit-Ribbit		
Penguin: Swim by using them wings that paddle-like flippers	}	Swim
Frog: Swim with their powerful hind legs, and flattened, streamlined body		
Penguin: Can sleep while floating in the water and standing up on the rocks	}	Sleep
Frog: May rest more throughout the day when the sun is hot		
Pig: Like to sleep snuggled up together in their nest		
Pig: Grill or Fry it	}	Eat
Fruit is healthy food		
Coconut: Refreshing with ice coconut water	}	Eat
Orange: Make Orange Juice		
Banana: Make Healthy Banana Smoothie		

```
D:\2\cpe121\8\mainpage_test_1.java
File Edit Tools Project Language Level Help
New Open Save Close Cut Copy Paste Undo Redo Find Compile Reset Run Test Javadoc Code Coverage

Animal.java
mainpage_test_1

1 public class mainpage_test_1{
2     public static void main (String[]args){
3         Penguin p = new Penguin();
4         Frog f = new Frog();
5         Pig pig = new Pig();
6         Fruit F = new Fruit();
7         Coconut c = new Coconut();
8         Banana b = new Banana();
9         Orange o = new Orange();
10
11        System.out.println(pig.sound());
12        System.out.println(p.sound());
13        System.out.println(f.sound());
14
15        System.out.println(p.howToSwim());
16        System.out.println(f.howToSwim());
17
18        System.out.println(p.howToSleep());
19        System.out.println(f.howToSleep());
20        System.out.println(pig.howToSleep());
21
22
23        System.out.println(pig.howToEat());
24        System.out.println(F.howToEat());
25        System.out.println(c.howToEat());
26        System.out.println(o.howToEat());
27        System.out.println(b.howToEat());
28    }
29 }
30
31
```

```
D:\2\cpe121\8\Animal.java
File Edit Tools Project Language Level Help
New Open Save Close Cut Copy Paste Undo Redo Find Compile Reset Run Test Javadoc Code Coverage

Animal.java
mainpage_test_1

1 interface Swimmable{
2     public String howToSwim();
3 }
4
5 abstract class Animal{
6     public abstract String sound();
7     public abstract String howToSleep();
8 }
9
10 interface Edible{
11     public String howToEat();
12 }
13
14 class Penguin extends Animal implements Swimmable{
15     public String howToSleep(){
16         return "Penguin: Can sleep while floating in the water and standing up on the rocks";
17     }
18     public String howToSwim(){
19         return "Penguin: Swim by using them wings taht paddle-like flippers";
20     }
21     public String sound(){
22         return "Penguin: Honk-Honk";
23     }
24 }
25
26 class Frog extends Animal implements Swimmable{
27     public String howToSwim(){
28         return "Frog: Swim with their powerful hind legs, and flattened, Streamlined body";
29     }
30     public String howToSleep(){
31         return "Froq : May rest more throughout the day when the sun is hot";
32     }
33     public String sound(){
34         return "Frog: Ribbit-Ribbit";
35     }
36 }
37
38 class Pig extends Animal implements Edible{
39     public String howToEat(){
40         return "Pig: Grill or Fry it";
41     }
42     public String howToSleep(){
43         return "Pig: Like to sleep snuggled up together in their nest";
44     }
45     public String sound(){
46         return "Pig: Oink-Oink";
47     }
48 }
49
50
51 class Fruit implements Edible {
52     public String howToEat(){
53         return "Fruit is healthy food";
54     }
55 }
56
57 class Coconut extends Fruit {
58     public String howToEat(){
59         return "Coconut: Refreshing with ice coconut water";
60     }
61 }
62
63 class Orange extends Fruit {
64     public String howToEat(){
65         return "Orange: Make Orange Juice";
66     }
67 }
68
69 class Banana extends Fruit {
70     public String howToEat(){
71         return "Banana: Make Healthy Banana Smoothie";
72     }
73 }
74
```

ส่วนของแม่



Quiz #08 Interface (27/04/2023)

2. จงทำการเขียน UML Class จากโปรแกรมที่กำหนดมาให้พร้อมทั้งเขียนโปรแกรมสำหรับใช้ในการทดสอบ

```
class Taxi extends Vehicle implements Passenger {
    public int speed = 0; public int gear = 0;
    public String howToPayMoney(){
        String text = "Taxi : Check the distances and cost from the meter. Pay money";
        return text; }
    public void changeGear(int newGear){
        gear = newGear; }
    public void speedUp(int increment){
        speed = speed + increment; }
    public void applyBrakes(int decrement){
        speed = speed - decrement; }
}
```

```
public abstract Vehicle {
    public abstract void changeGear(int a);
    public abstract void speedUp(int a);
    public void applyBrakes(int a);
}
```

```
class Bicycle extends Bike {
    public void printStates() {
        System.out.println("speed: " + speed + " gear: " +
        gear); }
}
```

```
public interface Passenger{
    public String howToPayMoney();
}
```

```
class Bike extends Vehicle implements Passenger {
    public int speed = 0; public int gear = 0;
    public void changeGear(int newGear){
        gear = newGear; }
    public void speedUp(int increment){
        speed = speed + increment; }
    public void applyBrakes(int decrement){
        speed = speed - decrement; }
}
```

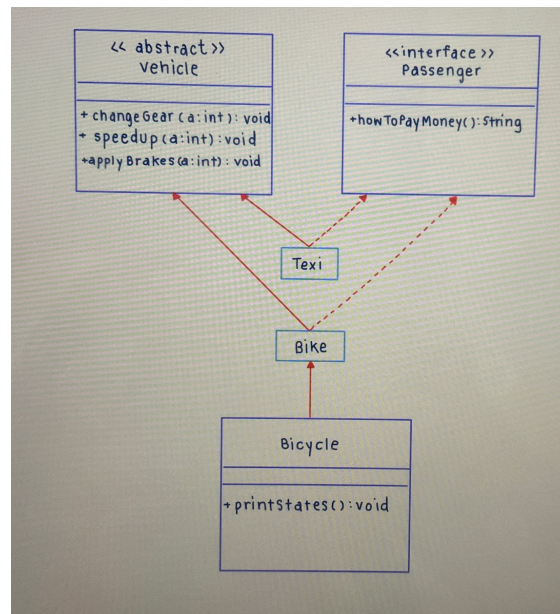
```
public class Quiz10 {
    public static void main(String[] args) {
        Taxi t = new Taxi( );
        System.out. print(t. howTo Pay money ());
        t. changeGear ( 1 );
        t. SpeedUp( 20 );
        t. applyBrakes( 10 );

        Bike b = new Bike ( );
        b. changeGear ( 2 );
        b. SpeedUp( 40 );
        b. applyBrakes( 10 );
    }
}
```

```
Bicycle B = new Bicycle( );
```

```
B. changeGear ( 3 );
B. SpeedUp( 80 );
B. applyBrakes( 10 );
B. printState ( );
```

UML Class





Quiz #08 Interface (27/04/2023)

2. จงทำการเขียน UML Class จากโปรแกรมที่กำหนดมาให้พร้อมทั้งเขียนโปรแกรมสำหรับใช้ในการทดสอบ

```
class Taxi extends Vehicle implements Passenger {
    public int speed = 0; public int gear = 0;
    public String howToPayMoney(){
        String text = "Taxi : Check the distances and cost from the meter. Pay money";
        return text; }
    public void changeGear(int newGear){
        gear = newGear; }
    public void speedUp(int increment){
        speed = speed + increment; }
    public void applyBrakes(int decrement){
        speed = speed - decrement; }
}
```

```
public abstract Vehicle {
    public abstract void changeGear(int a);
    public abstract void speedUp(int a);
    public void applyBrakes(int a);
}
```

```
class Bicycle extends Bike {
    public void printStates() {
        System.out.println("speed: " + speed + " gear: " +
        gear); }
}
```

```
public interface Passenger{
    public String howToPayMoney();
}
```

```
class Bike extends Vehicle implements Passenger {
    public int speed = 0; public int gear = 0;
    public void changeGear(int newGear){
        gear = newGear; }
    public void speedUp(int increment){
        speed = speed + increment; }
    public void applyBrakes(int decrement){
        speed = speed - decrement; }
}
```

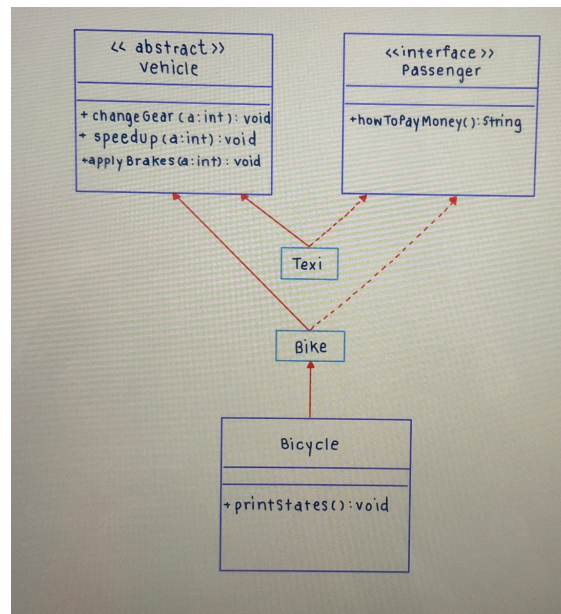
```
public class Quiz10 {
    public static void main(String[] args) {
        Taxi t = new Taxi( );
        System.out. print(t. howTo Pay money ());
        t. changeGear ( 1 );
        t. SpeedUp( 20 );
        t. applyBrakes( 10 );

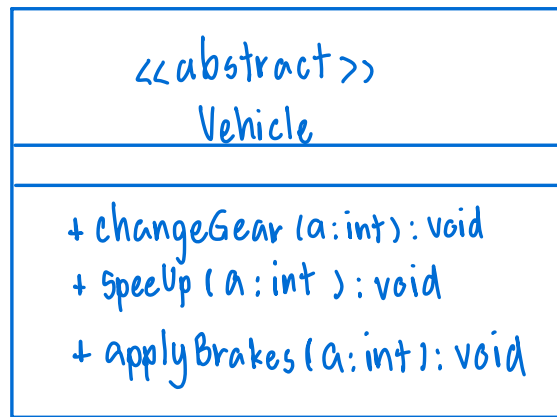
        Bike b = new Bike ( );
        b. changeGear ( 2 );
        b. SpeedUp( 40 );
        b. applyBrakes( 10 );
    }
}
```

```
Bicycle B = new Bicycle( );
```

```
B. changeGear ( 3 );
B. SpeedUp( 80 );
B. applyBrakes( 10 );
B. printState ( );
```

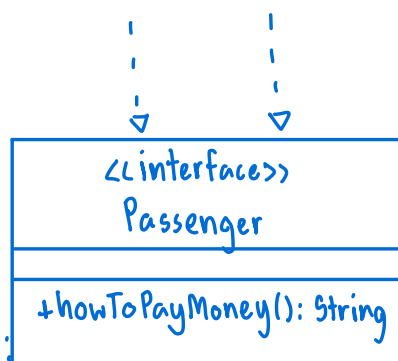
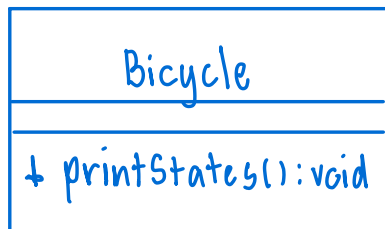
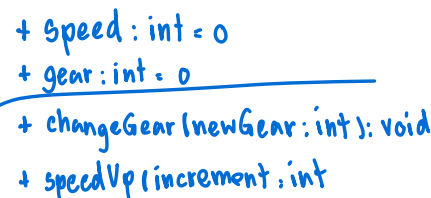
UML Class





Bike

Taxi



```

Taxi taxi = new Taxi();
Bike bike = new Bike();
Bicycle bicycle = new Bicycle();
System.out.println(taxi.howToPayMoney());
taxi.changeGear(1);
taxi.speedUp(1);
taxi.applyBrakes(1);
bike.changeGear(1);
bike.speedUp(1);
bike.applyBrakes(1);
bicycle.printStates();

```